1.

HW3_1.cpp

```cpp
#include<iostream>
using namespace std;

int main()
{
    int time,cost;
    char type;
    cout << "Input type of vehicle: car (c) bus (b) truck (t):" ;
    cin >> type;
    cout << "Input the number of hours:" ;
    cin >> time;

    if (type=='c')
    cost = 2*time;
    else if(type=='b')
    cost = 3*time;
    else if(type=='t')
    cost = 4*time;

    cout << "You owe us:$" << cost << endl << "Thank you for parking with us";

    return 0;
}
```

```
Input type of vehicle: car (c) bus (b) truck (t):c
Input the number of hours:6
You owe us:$12
Thank you for parking with us
--------------------------------
Process exited after 11.74 seconds with return value 0
請按任意鍵繼續 . . .
```

2.

HW3_2.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
    int number;
    float price,discount,total;

    cout << "Enter number of items:";
    cin >> number;
    cout << "Enter unit price:";
    cin >> price;
    price *= number;

    if (number>0 && number<10)
        discount = 0;
    else if (number>9 && number<50)
        discount = price*0.03;
    else if (number>49 && number<100)
        discount = price*0.05;
    else if (number>99)
        discount = price*0.1;

    total = price-discount;

    cout << "Subtotal:" << price << endl;
    cout << "Discount:" << discount << endl;
    cout << "Total:" << total;
    return 0;
}
```

```
Enter number of items:30
Enter unit price:10
Subtotal:300
Discount:9
Total:291
--------------------------------
Process exited after 2.992 seconds with return value 0
請按任意鍵繼續 . . .
```

3.

HW3_3.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
    int score;
    char grade,modifier;

    cout << "Enter a score between 0 and 100:";
    cin >> score;

    switch(score/10)
    {
        case 10: grade = 'A';
            break;
        case 9: grade = 'A';
            break;
        case 8: grade = 'B';
            break;
        case 7: grade = 'C';
            break;
        case 6: grade = 'D';
            break;
        default: grade = 'F';
    }

    switch(score%10)
    {
        case 9:
        case 8:
            modifier = '+';
            break;
        case 0:
        case 1:
        case 2:
        case 3:
            modifier = '-';
            break;
    }

    if (score==100)
        modifier = '+';
    else if (score<60)
        modifier = ' ';

    cout << "Score:" << score <<endl;
    cout << "Grade:" << grade << modifier;

    return 0;
}
```

```
Enter a score between 0 and 100:78
Score:78
Grade:C+
--------------------------------
Process exited after 2.516 seconds with return value 0
請按任意鍵繼續 . . .
```

4.

HW3_4.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
    int oper;
    float num1,num2,remain;

    cout << "Menu:" << endl;
    cout << "1.Addition" << endl;
    cout << "2.Subtraction" << endl;
    cout << "3.Multiplication" << endl;
    cout << "4.Division" << endl;

    cout << "Select one of the operations (1,2,3,4):";
    cin >> oper;

    if(oper<1||oper>5)
    {
        cout << "Wrong operation! The program terminates.";
        return 0;
    }
    cout << "Enter the value of the first number:";
    cin >> num1;
    cout << "Enter the value of the second number:";
    cin >> num2;

    if (oper==4 && num2==0)
    {
        cout << "Error division by zero. The program terminates.";
        return 0;
    }

    switch (oper)
    {
        case 1: cout << num1 << "+" << num2 << "=" << num1+num2;
            break;
        case 2: cout << num1 << "-" << num2 << "=" << num1-num2;
            break;
        case 3: cout << num1 << "*" << num2 << "=" << num1*num2;
            break;
        case 4: cout << num1 << "/" << num2 << "=" << num1/num2;
            break;
    }

    return 0;
}
```

```
Menu:
1.Addition
2.Subtraction
3.Multiplication
4.Division
Select one of the operations (1,2,3,4):1
Enter the value of the first number:2.7
Enter the value of the second number:3.5
2.7+3.5=6.2
--------------------------------
Process exited after 5.166 seconds with return value 0
請按任意鍵繼續 . . .
```